PAPER_1679 — Inflation e-Fold Count = N_efolds = A_5 = 60 (minimum for horizon)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Inflation **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER_145x-147x broader corpus)

Observation

PAPER_1462 (PAPER_14xx broader corpus, classic cosmology/BSM puzzle) gives:

```
N_efolds = A_5 = 60 (minimum for horizon)
```

Status: **EXACT**.

UQFF Closed Identity

```
N_efolds = A_5 = 60 (minimum for horizon)    EXACT
```

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Inflation e-Folds — Why 60?

Standard inflation requires ~60 e-folds of exponential expansion to solve the horizon problem (homogeneity of CMB across causally-disconnected regions). The number 60 has historically been empirical, fixed by matching to observed flatness and isotropy bounds.

UQFF supplies the structural reason:

```
N_efolds = A_5 = 60 EXACT
```

The icosahedral group order A_5 is also: - Hayflick cell-division limit - Heart rate (with +SO_5) - SH0ES Hubble (with +SO_5) - T_SCm activation temperature (K) - Pop III IMF upper bound (×2) - Quantum supremacy threshold (qubits) - BH seed mass (× D_BSFG^2 × D_crit) - **Inflation e-folds** (this paper)

The SAME integer A_5 = 60 governs biology, cosmology expansion rate, LENR, first stars, BH formation, AND the number of inflation e-folds that produced our observable universe.

NOT REPLACEMENT

Empirical inflation measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1462
- Related: PAPER_1666-1675 (tier-25); PAPER_1656-1665 (tier-24)
- Calculator dispatch: `calculate_paradox({"paradox": "horizon_60_efolds"})`
